HYEJIN PARK

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EDUCATION

- **Rensselaer Polytechnic Institute (RPI)** *Expected May 2029*
  Ph.D. in Computer Science (Advisor: Prof. Tomek Strzalkowski) Troy, New York, US
  - **Affiliations:** Language and Cognition Artificial Intelligence (LACAI) Lab & Tetherless World Constellation (TWC)
  - **Focus:** Trustworthy NLP, LLM Safety & Watermarking, Semantic Reasoning, Human-AI Collaboration
- **Yonsei University** *Aug 2022*
  Master's Degree in Computer Science (Advisor: Prof. Yo-sub Han) Seoul, Republic of Korea
- **Hansung University** *Aug 2019*
  B.S. in Industrial Management & Computer Engineering (Advisor: Prof. Kitae Hwang) Seoul, Republic of Korea

SELECTED PUBLICATIONS

- **A Linguistics-Aware LLM Watermarking via Syntactic Predictability**
  S. Park, **H. Park**, H. Ahn, Y. Han
  Accepted to ACL 2026 2026
- **WaterMod: Modular Token-Rank Partitioning for Probability-Balanced LLM Watermarking**
  S. Park, **H. Park**, H. Ahn, Y. Han
  AAAI Conference on Artificial Intelligence (AAAI-26), **Oral Presentation** 2026
- **KHG-Aclair: Knowledge Hypergraph-Based Attention with Contrastive Learning**
  H. Park, T. Lee, K. Kim
  Journal of Computing Science and Engineering (JCSE), 18(3), 169-180. 2024
- **TRIO: An Entity Retrieval Method Using Entity Embedding and Topic Modeling**
  H. Park, D. Woo, S. Park, K. Kim
  Journal of Computing Science and Engineering (JCSE), 18(1), 36-46. 2024
- **An Implementation of Smart Gardening Using Raspberry Pi and MQTT**
  K. Hwang, **H. Park**, J. Kim, T. Lee, I. Jung
  The Journal of the Institute of Internet, Broadcasting and Communication, 18(1), 151-157 2018

VOLUNTEER & SCHOLARSHIP EXPERIENCE

- **ACL 2026 Student Volunteer Program** *2026*
  Student Volunteer & Registration Waiver Recipient Association for Computational Linguistics
  - Selected as a student volunteer and service commitment.
- **AAAI-26 Student Scholar Volunteer Program** *2026*
  Travel Scholarship Recipient ($1,500) Association for the Advancement of Artificial Intelligence
  - *Details:* Selected based on research contribution (AAAI-26 Oral Presentation) and service commitment.

RESEARCH EXPERIENCE

- **Language and Cognition Artificial Intelligence (LACAI) Lab, RPI** *May 2025 - Present*
  Graduate Research Assistant Troy, NY, US
  - **Project:** DARPA ECOLE / ARCLIGHT Program
  - **VLM Reasoning & Semantic Blinding:** Engineered a "semantic blinding" pipeline to strictly evaluate object affordance and physical reasoning in Vision-Language Models (VLMs), preventing reliance on prior text labels.
  - **Multimodal Representation Learning:** Developed a large-scale (620K+ samples) video-affordance annotation pipeline using VLMs and designed a contrastive learning framework to fine-tune backbones for zero-shot affordance discovery.

- **Korea Electronics Technology Institute (KETI)** *Sep 2022 - May 2024*
  Researcher Seoul, Republic of Korea
  - **Maritime Digital AtoN System (Sponsor: Ministry of Oceans and Fisheries):** Developed analysis algorithms for quality management using integrated big data from smart navigational markers, resulting in a first-author conference publication.
  - **Multi-Agent Simulation (Sponsor: Ministry of Science and ICT):** Developed a PII de-identification model for legal/traffic data and automated OpenDrive file pattern identification for road transformation, leading to a first-author journal publication.

- **Yonsei University** *Apr 2021 - Aug 2022*
  Graduate Research Assistant (Advisor: Prof. Yo-Sub Han) Seoul, Republic of Korea
  - **Neuro-Symbolic AI for Fact-Checking (Sponsor: Ministry of Science and ICT):** Developed efficient search algorithms for query-based knowledge graphs and converted retrieved knowledge into Neuro-Symbolic Graphs to logically support facts from raw text, publishing a first-author journal article.

Leadership & Teaching Experience

- **Rensselaer Polytechnic Institute (RPI)** *Fall 2024 - Summer 2025*
  Teaching Assistant Troy, NY, US
  - **Courses:** Data Structures (CSCI 1200), Python Bridge (CSCI 1100), Data & Society (CSCI 4370).
  - Mentored undergraduate students in core CS concepts, led programming lab sessions, and facilitated discussions.

- **Yonsei University** *Fall 2020*
  Teaching Assistant Seoul, Republic of Korea
  - **Course:** Data Structures (CSI 2103) – *English-only Class*.
  - Guided students and communicated complex algorithmic concepts effectively in a fully English-spoken academic environment.

- **Tech Start-up & Engineering Projects** *Sep 2017 - Dec 2019*
  Team Lead & Co-Founder (SK Start-Up Club) Seoul, Republic of Korea
  - **Team Leadership:** Founded a student IoT startup and directed multiple cross-functional teams to develop smart monitoring systems (Greeni) and dashboard authoring tools (YellowPeach).
  - **Key Achievements:** Demonstrated strong project management and product delivery, winning **1st Prize** in the SK Start-up Competition and **2nd Prize** in the Embedded Software Competition.

Selected Awards & Honors

- **Finalist, AI Grand Challenge** *Jun 2021*
  Nationwide R&D Competition Ministry of Science and ICT
  - *Project:* Mathematical Reasoning AI - Model Providing Solution Steps for Text-based Math Problems.

- **Gold Prize, Open Source SW Developer Competition** *Dec 2019*
  Nationwide Development Competition Ministry of Science and ICT

- **Undergraduate Distinguished Achievement Award** *Aug 2019*
  University-level Award Hansung University

- **2nd Prize, 16th Embedded Software Competition** *Dec 2018*
  Nationwide Development Competition Ministry of Trade, Industry, and Energy

- **2nd Prize, K-Hackathon (Digital Solution Challenge)** *Nov 2018*
  Nationwide Development Competition Ministry of Science and ICT

Technical Skills

- **Languages:** Python, C/C++, Java, SQL, JavaScript
- **AI/ML & HPC:** PyTorch, Hugging Face Transformers, vLLM, CUDA, MPI, Multi-GPU Computing
- **Tools & Platforms:** Git, Docker, Linux, LaTeX, OpenCV, AiMOS Supercomputing Environment